What's new

Satz vom Nullprodukt

Terence Tao · 14 February 2022

https://terrytao.wordpress.com/2022/02/14/satz-vom-nullprodukt/

tao2typ · converted to Typst 2026-10-14 (UTC)

As I have mentioned in some recent posts, I am interested in exploring unconventional modalities for presenting mathematics, for instance using media with high production value. One such recent example of this I saw was a presentation of the fundamental zero product property (or domain property) of the real numbers – namely, that $ab = 0$ implies $a = 0$ or $b = 0$ for real numbers a, b – expressed through the medium of German-language rap:

EDIT: and here is a lesson on fractions, expressed through the medium of a burger chain advertisement:

I'd be interested to know what further examples of this type are out there.

SECOND EDIT: The following two examples from Wired magazine are slightly more conventional in nature, but still worth mentioning, I think. Firstly, my colleague at UCLA, Amit Sahai, presents the concept of zero knowledge proofs at various levels of technicality:

Secondly, Moon Duchin answers math questions of all sorts from Twitter:

Comments (56)

 **Anonymous** · 14 February 2022

Is the product operation the simplest (in some sense) binary operation which satisfy the product property over the reals?

>  **Terence Tao** · 14 February 2022
>
> In a sense, yes! The set $\{(a,b) : a = 0 \text{ or } b = 0\}$ is a (reducible) algebraic curve of degree 2, so the lowest degree non-trivial polynomial $P(a,b)$ of two variables for which one can have the zero product property $P(a,b) = 0 \implies a = 0 \text{ or } b = 0\}$ is indeed just the multiplication operation $P(a,b) = ab$.

 **Adam Levine** · 14 February 2022

I'd recommend my song to anyone looking to learn about Heegaard Floer homology.

 **Edgawliet** · 14 February 2022

What about a Spanish song?

 **Jochen Voss** · 14 February 2022

How about this one? https://www.youtube.com/watch?v=uFvokQUHh08

 **kapple19** · 14 February 2022

Many of acapellascience's [https://youtube.com/c/acapellascience] parodical music videos make references to and sometimes present various levels of mathematics.
Additionally, from a musical standpoint his harmonies are of a richly and high quality.

 **Aditya Guha Roy** · 14 February 2022

You might be aware of this but let me share it here, just in case you didn't watch this earlier. https://www.youtube.com/watch?v=IUTGFQpKaPU
Here Professor Po-Shen Loh extends the joke-conversation "get real, be rational" between i and pi to elaborate to draw light on Euler's identity.

 **Aditya Guha Roy** · 14 February 2022

Also there is a nice book titled "What's your angle" which draws light on Pythagoras's theorem by means of a story written in a tone which is meant for kids (but some grown-ups may also find it fun to read, like I did).

 **Aditya Guha Roy** · 14 February 2022

There is also a series titled "Manga guide to ..... " which describes certain math topics through comics.
For instance: there is the manga guide to linear algebra , the manga guide to calculus, the manga guide to relativity, the manga guide to statistics, etc..

 **Aditya Guha Roy** · 14 February 2022

Also there are some comic strips which concern some mathematical results. I remember seeing one such example about dominated convergence theorem, but sadly I can't find it anymore, and my old computer had it saved, but now it's gone and I certainly don't have it in my new computer anymore.

 **Aditya Guha Roy** · 14 February 2022

I have this habit of playing an online game by the name agar.io and there I customize the skins of my agar bubble by writing pieces of math theorems and sometimes even complete proofs (proof without words form).
Recently I found two other players using customized skins with mathematical things written on it, and it feels good to find people who think alike about this way of promoting mathematics :D .

 **Anurag Bishnoi** · 15 February 2022

I was involved with making this video that explains Ramsey numbers via game of thrones: https://www.youtube.com/watch?v=WN_tQWIkeV0&ab_channel=anuragbishnoi
I would have loved to make a longer video, but this was for the Veritasium contest and there was max limit of 1 min.

>  **Aditya Guha Roy** · 15 February 2022
>
> Yeah I remember seeing this before; I shared it with some of my friends who are very much up-to-date with the game of thrones movies (although I never watched them, but I have friends who watched them and follow the series closely) but who did not know this thing before. One of them exclaimed that it was like magic.

 **Anonymous** · 15 February 2022

It doesn't entirely fit your request, however it still provokes a good laugh ;-) https://www.youtube.com/watch?v=2un9rO2ZF4g

 **Aditya Guha Roy** · 15 February 2022

I just asked some of my friends, and one of them said that there are some YouTube videos which show how mathematics can be demonstrated through Lego bricks. So, you may like to find that out.

 **Aditya Guha Roy** · 15 February 2022

For a good laugh you may consider watching this. https://www.youtube.com/watch?v=SyD4p8_y8Kw

 **Michael Hunter** · 15 February 2022

I think a good video would be one on a representation domain for a subject area. For example
[three integrals] Calculus can equal L*W*H Geometry
I think it may be representation that may cause an issue with people trying to learn a subject. If people learning, understand the domain of representation (identities/formulas) within a subject area, they may be better suited to tackle said subject. Then as the become better mathematicians can understand how the subjects identities can overlap and make math more powerful.

 **Roberto D'Autilia** · 15 February 2022

I imagine that you all know this

 **José Hdz** · 15 February 2022

The burger chain advertisement reminded me of the recent beer ad in which Professor Ken Ono participated:

 **Anonymous** · 15 February 2022

Equations song

Adam Levine · 15 February 2022

These other math songs are so poor compared to mine. (See above.)



Tom Weidig · 15 February 2022

The German rap is very cool! And the German wording is also well done. I sent it to my teenage nieces!;-)

Florian · 16 February 2022

There is a German channel, called DorFuchs, which is dedicated to mathematical songs and math communication, see https://www.youtube.com/c/DorFuchs
This channel contains songs ranging from the binomial formulas and polynomial division to the irrationality of pi.

Olof Sisask · 16 February 2022

My students mentioned DorFuchs in the context of Calculus just yesterday! In particular, the lyrics you can get out of "cosinus", "sinus" and "minus" coupled with differentiation and integration:
Much less entertaining, but hopefully useful nonetheless, is an ongoing experiment of mine together with a couple of others to create an app to help students with Calculus problem solving: https://www.dogl.app. I'll have to look at adding German-language math music to the app — clearly the pedagogical medium of the future!

Helene · 16 February 2022

AGAIN，I cannot see the video you uploaded...what a pity...

Satz vom Nullprodukt | Blog.CarlRobitaille.org · 16 February 2022
[...] Satz vom Nullprodukt [...]

Liewyee · 16 February 2022

sorry，maybe nothing about this blog，but I really want to know，what do you mean by SPENDING in spending symmetry?

mgflax · 17 February 2022

Suppose you wished to prove something about the relationship between two points on a circle. By symmetry, you can choose the first point — without any loss of generality — to be a convenient value, e.g. (1,0). But now you've "spent" the symmetry and you can't also assume that the second point is also at a particularly-convenient value.

Diego · 17 February 2022

This song is also in german and it's about the geometric series convergence.

Th3o_r∈M · 17 February 2022

I am the author of the "Satz vom Nullprodukt" german math rap video.
Honestly I couldnt believe that it was linked here. This is a great honor for me! Thank you so much!
🙏
I will add some english subtitles asap.
As a teacher i felt responsible to create a different approach for my audience in the age of 16+. I try to mix good entertainment in modern HipHop Beats with education realized by mathematical Raps. In an anonymous online feedback one student wrote that he/she enjoys mathematics and having fun for the first time after 10 years of school.

Most of the students in school need some motivation in math and modern & entertaining music can help with that. I know that they hear my math song in their freetime as well. ;)
FYI: My next song will be released on Friday, 18th Feb. Title: Symmetry

Anonymous · 19 February 2022

[I don't know why WordPress does not show my comment. Let me try one more time.]
You mentioned this one in a blogpost more than 10 year ago: Moebius Transformations Revealed

> **J.** · 19 February 2022
>
> By the way, how are the viewpoints of the mentioned speech evolved over the years? :-)
> Recently, there have been videos from 3Blue1Brown.
> And your friend Charlie Fefferman's talks about infinite series and the Kakeya's Needle problem on Numberphile.

> **J.** · 19 February 2022
>
> And your talk on Numberphile about 'Cheating Strategically'.

> **J.** · 19 February 2022
>
> And your talk about 'Cheating Strategically' on Numberphile.

> **J.** · 19 February 2022
>
> It seems that the examples above (by Charlie Fefferman and the Mobius one) are not of "unconventional modalities" though.
> Do you think unconventional modalities are always possible for presenting mathematics at any level? What in general would you classify as conventional?



Jas, the Physicist · 19 February 2022

I've written a freestyle a while ago about S1 as a complex manifold but the wordplay was too vague so I disregarded it.

> **Anonymous** · 19 February 2022
>
> Maybe because S^1 isn't a complex manifold



> **Jas, the Physicist** · 20 April 2022
>
> What isnt a complex manifold? S1?
> What. Did I miss something?
> You can properly embed it into C with a smooth, not even just smooth, but holomorphic atlas.
> Get tf o.
> I need harder questions than this.
> Okay you're going to ask me which function then.
> And I say the exponential function.
> Qed.

> **Anonymous** · 20 April 2022
>
> There is no injective analytic map from S^1 to the disc. Maybe you're thinking of S^2.



> **Jas, the Physicist** · 21 April 2022
>
> S1 is a complex manifold. That's the point I'm trying to make. Also what disc are you talking about and how is it related?

Jas, the Physicist · 21 April 2022

Oh I read about the neighborhoods required for local homemomorphsm and I see what you mean by D now. But I can see S1 properly embedded in C so how is it not a manifold? I can use the level sets of…

Anonymous · 21 February 2022

https://www.youtube.com/channel/UCIqiLefbVHsOAXDAxQJH7Xw

Anonymous · 22 February 2022

Eulers number is irrational – proof & song

Swag Eagle · 22 February 2022

“Finite simple group of order 2” is epic but maybe not in the scope of what you are looking for. I presume, however, that what follows is in the scope (in French):

Anonymous · 24 February 2022

The rap video has been removed:

Video unavailable
This video is no longer available because the YouTube account associated with this video has been terminated.

Th3o_r∈M · 24 February 2022

I am working at the issue of the removed Rap video. I will inform you when it is back online. In the worst case i will send a new link.. But hopefully it will work in a little while.
Kind regards
Th3o_r∈M

Th3o_r∈M · 24 February 2022

Update:
The linked Rap Video is online again.
The Youtube Team apologize for making a mistake here. Also from my side: I am sorry for the inconvenience.
If you like you can also check my website. I will release some teaching materials that will be useful for students to develop the topics with my Rap musicvideos. (Some teachers worked already successfully with my music videos.)
My next math rap song is already recorded and will be released in 4 weeks.
Thank you for your support. <3
Kind regards
Th3o_r∈M

Anonymous · 24 February 2022

Will the ICM 2022 still going to happen in Russia?
[As per this notification, there will be a formal announcement about this by the IMU shortly. -T]

Anonymous · 28 February 2022

Why keep defaulting to these one-off songs? They take
up so much time and effort to come up yet they barely
explain one little concept.
All of these are just “cute” ways to help students remember.
They don’t foster any understanding of the underlying concept.
Instead of coming up with ad-hoc attempts, a better way to captivate students would be the following.
1) Instructors of advanced math courses are really good
at putting students in the “weeds.” Give them
a birds eye view of what math can do in neighboring
fields. (Using Freeman Dyson’s analogy here.)
2) Interweave lessons with storie and history.
Math is developed by real people with real stories.
They serve as good anchor points and they will remember better.
Turn math into stories, not lessons.
3) How about some zest when teaching?
Some educators have as much enthusiasm as Walmart
employees. Eddie Woo at Youtube is great because he’s so damn
enthusiastic!
His channel here: https://www.youtube.com/c/misterwootube

Liewyee · 1 March 2022

Professor Tao,
can we use mathmetic method to demonstrate that WE DON’T KNOW SOME PARTICULAR INFORMATION?

> **Anonymous** · 1 March 2022
>
> This is not how mathematics work. Just use your reasoning: how do we know what we don’t know?

> **Anonymous** · 2 March 2022
>
> For instance by showing that certain proposition is undecidable in some given axiomatic system (e.g. the continuum hypothesis was shown by Godel and Cohen to be independent, and consequently undecidable in the standard ZFC axiomatic set theory)



Aditya Guha Roy · 15 March 2022

I think you know this, but since nobody mentioned: there is a nice weblog called “Math with bad drawings”, although it doesn’t teach anything new but it gives a fun way of looking at known things; I wonder why you haven’t put it up, even though it is mentioned in your blog-roll column.

Joseph O'Rourke · 1 April 2022

Matt Parker (Stand-Up Maths) makes consistently informative videos. Here’s one that touched on my work, which he entitled “How many 3D nets does a 4D hypercube have?”: https://www.youtube.com/watch?v=Yq3P-LhlcQo .

Anonymous · 10 June 2022

Everyone have a look at this music video, on a great mathematician.:

Anonymous · 10 June 2022

This is vulgar.